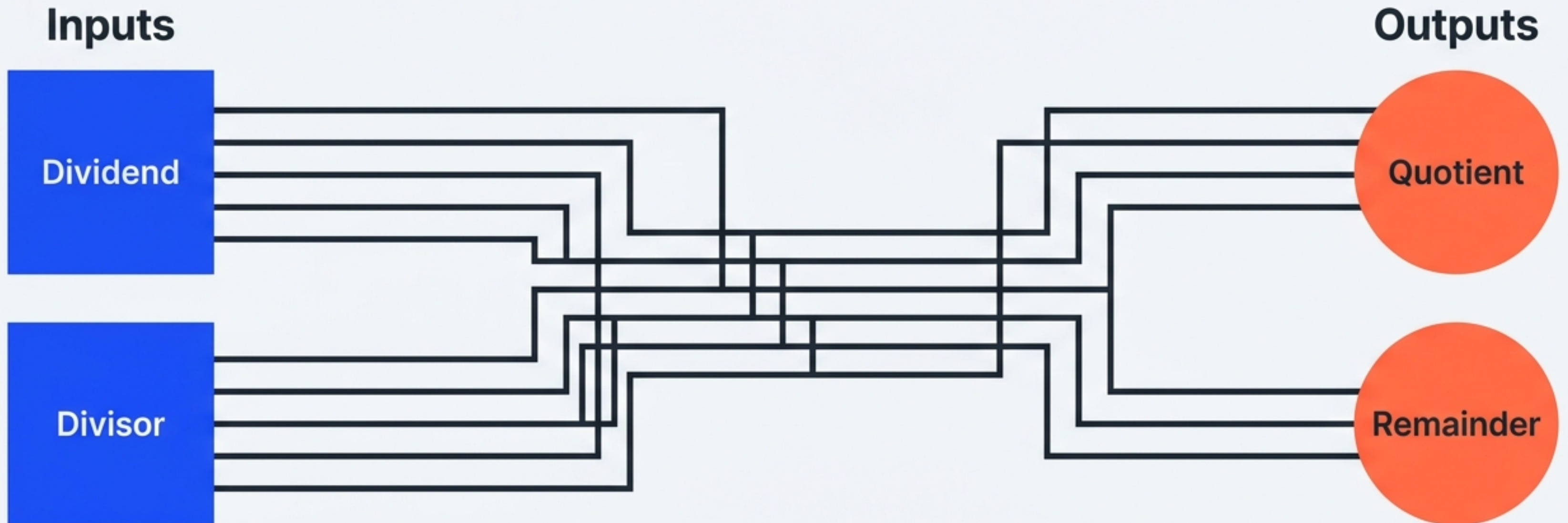
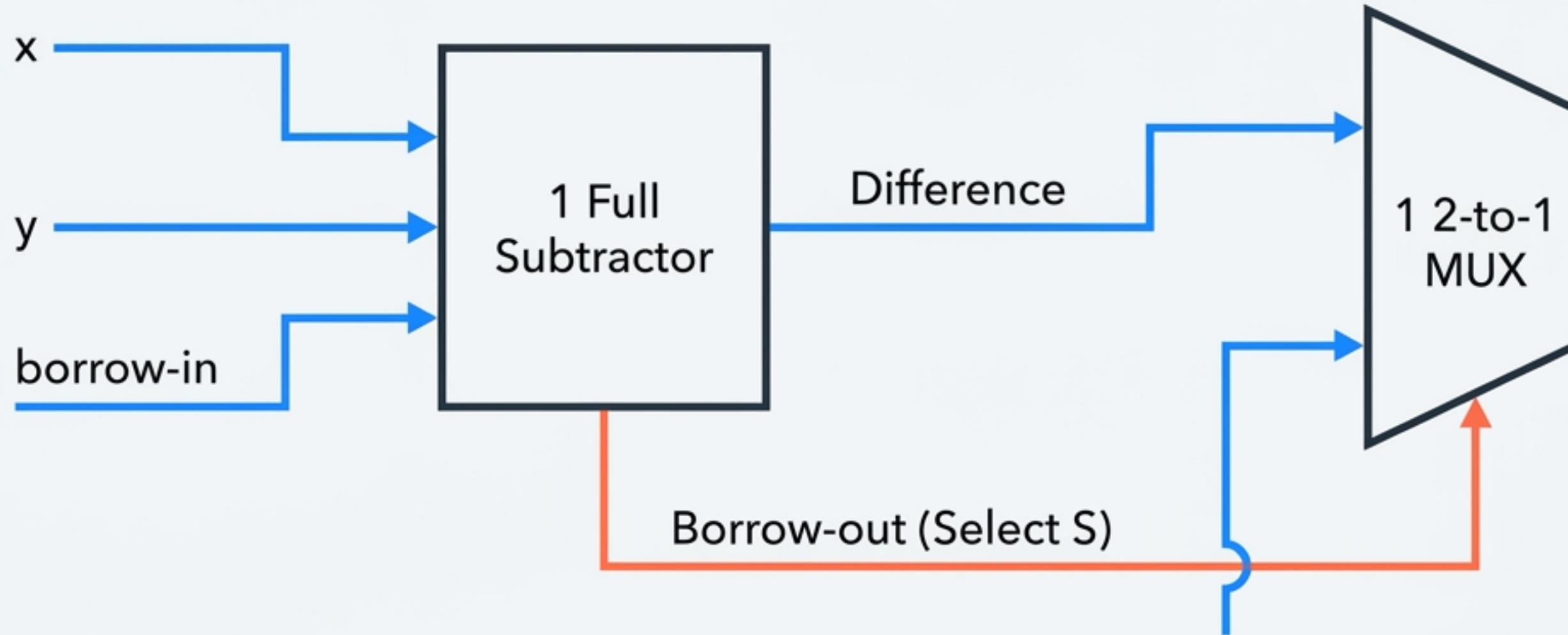


Combinational Dividers

$$\text{Dividend} = \text{Divisor} * \text{Quotient} + \text{Remainder}$$



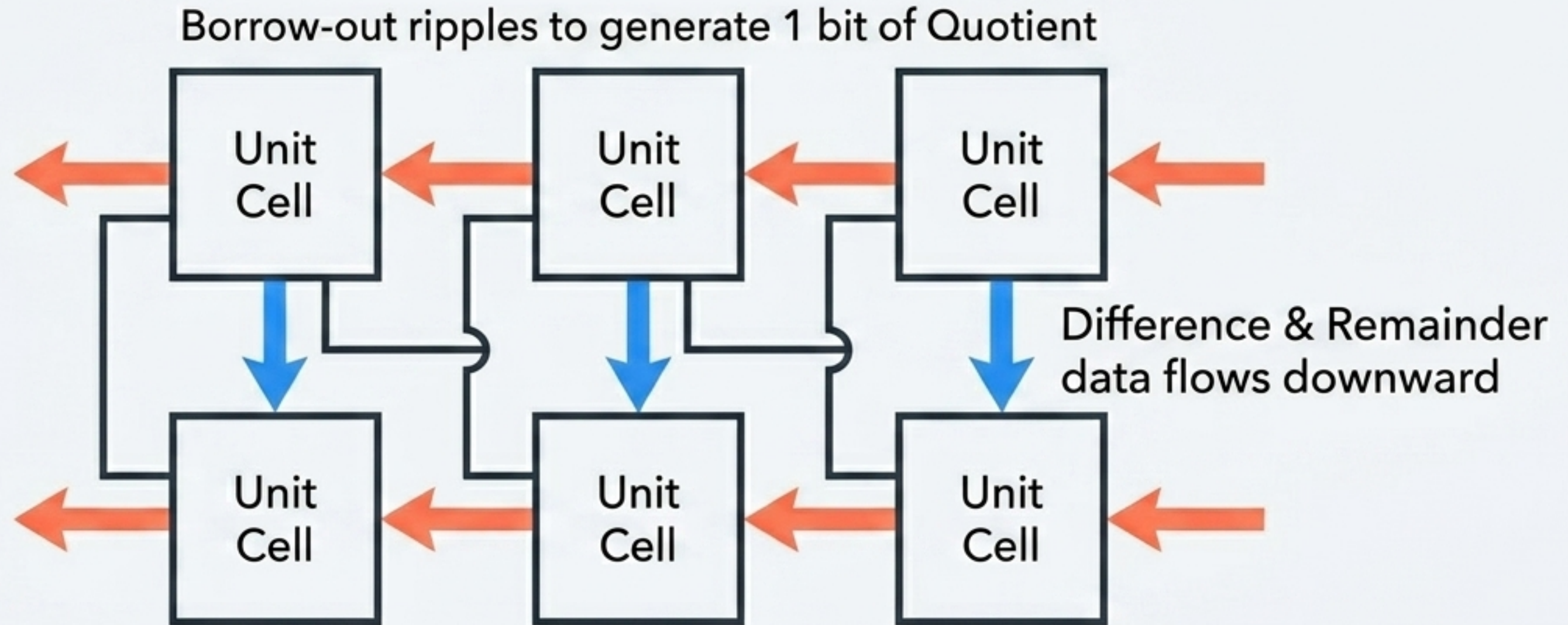
Unit Cell Architecture



If Borrow-out = 0: MUX passes Difference.

If Borrow-out = 1: MUX passes original Minuend.

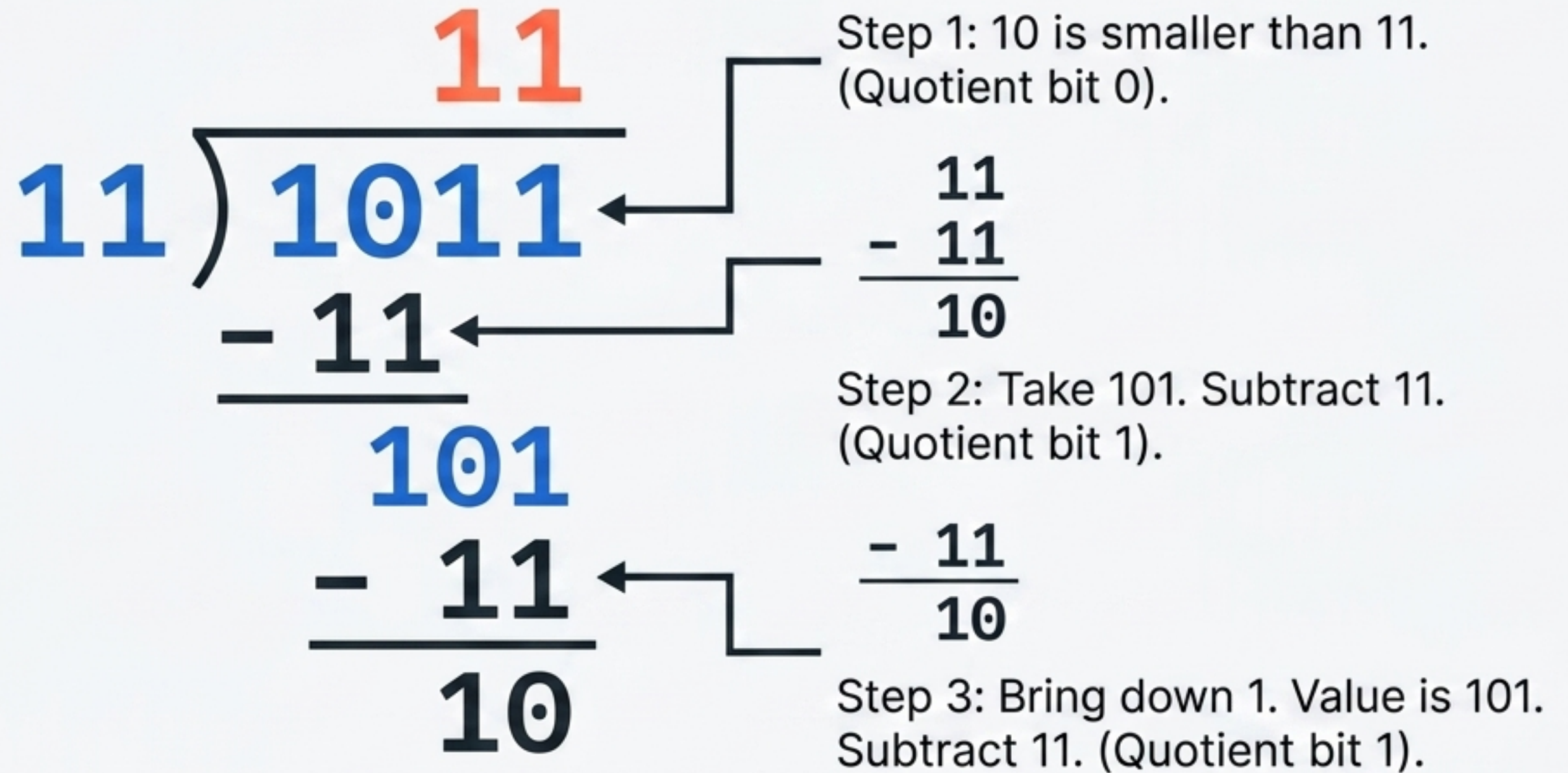
The Array Structure



Mechanism: Repetitive shifting and conditional subtraction.

Action: Divisor conceptually shifts right by 1 bit against the Dividend as data flows downward through 1 row of the array.

Numerical Execution: 1011 / 11



Result: Quotient = 11, Remainder = 10